GradTDDFT: A Differentiable Time-Dependent Density Functional Theory Framework for Training Neural Network Exchange-Correlation Functionals

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GradTDDFT is a JAX-based framework for differentiable time-dependent density functional theory (TDDFT) with neural exchange-correlation (XC) functionals. The method separates a fixed molecular data layer—basis functions, quadrature grids, one-electron integrals, and electron-repulsion integrals—from the differentiable Kohn–Sham SCF and TDDFT response layers. This separation gives a well-defined computational graph from neural XC parameters to self-consistent orbitals, TDA/Casida response matrices, excitation energies, and oscillator strengths. The central technical element is an energy-consistent local response tensor: the XC potential and f_{xc} kernel are obtained as first and second derivatives of the same learned local energy density by JAX automatic differentiation. The neural functional uses a coefficient-basis architecture in which pointwise density, gradient, kinetic-energy-density, local exact-exchange, and optional MP2/PT2 descriptors produce mixing fields over fixed semilocal, HF, and MP2/PT2 channels. We formulate the resulting ground-state and excited-state objectives together with the corresponding computational evaluation protocol.

Keywords: differentiable programming, JAX, TDDFT, neural exchange-correlation functional, excited states

I. INTRODUCTION

Density functional theory (DFT) and its linear-response time-dependent extension (LR-TDDFT) provide efficient routes to ground-state and excitation energies in molecular electronic structure.[1–4] Their predictive accuracy is ultimately controlled by the exchange-correlation (XC) approximation, which must provide not only an energy but also the derivatives that enter the self-consistent potential and the response kernel. In Kohn–Sham DFT, the first functional derivative of the XC energy determines the XC contribution to the SCF potential; in adiabatic LR-TDDFT, the second derivative enters the Casida response matrices that govern excitation energies and oscillator strengths. The XC functional therefore determines the ground-state SCF reference that underlies subsequent excited-state response calculations.

Finding a suitable XC functional is therefore central to reliable ground-state SCF and excited-state response calculations. Neural network XC functionals provide a flexible route to learn such functional forms, including nonlocal XC structure, from accurate reference data.[5,6] For such excited-state targets, however, fitting ground-state energies alone is not enough: the functional must also produce stable self-consistent densities, useful frontier orbital gaps, and a response kernel consistent with the learned XC energy. This requirement naturally leads to differentiable training, in which the functional is optimized through the electronic-structure calculation that will ultimately use it.[7] A recent overview of differentiable DFT emphasizes that this changes the role of the KS equations: the functional is not only evaluated inside a self-consistent DFT calculation, but trained through the differentiable SCF map, with gradients from observables propagated back to the model parameters while the governing electronic-structure equations remain in the loop.[8] For TDDFT, the corresponding differentiable program must extend beyond the ground-state SCF fixed point to include response-kernel construction, TDA/Casida eigenvalue problems, excitation energies, oscillator strengths, and spectrum-level objectives. Differentiable training is therefore important because it can place losses on the quantities actually used to judge excited-state predictions, rather than only on pre-response ground-state energies.

This training view is supported by a broader set of differentiable algorithms in molecular simulation, semiempirical electronic structure, quantum chemistry, and materials calculations.[9–15] For the present problem, the most relevant progress lies in differentiable DFT and XC functional construction. DQC and D4FT expose differentiable DFT workflows for learning or optimizing density functionals,[16,17] GradDFT provides a ground-state framework for machine-learning enhanced density functionals,[18] and jax-xc makes conventional Libxc energy-density forms available as differentiable functional channels.[19] In parallel, neural XC models have explored more flexible functional forms, including nonlocal and coefficient-basis descriptions beyond a fixed semilocal ansatz.[5,6,20] Together these works show how XC functionals can be represented, evaluated, and trained inside self-consistent DFT calculations. The missing piece for excited states is an end-to-end differentiable TDDFT formulation in which the neural XC energy, SCF potential, response kernel, and excited-state observables belong to the same differentiable computational graph.

We present GradTDDFT to address this gap by extending GradDFT from ground-state neural XC functional modeling to differentiable linear-response TDDFT training. The neural XC model is written as a coefficient-basis functional with trainable semilocal, HF, and optional MP2/PT2 channels. Its semilocal basis can be built from jax-xc energy-density components, so conventional LDA, GGA, and meta-GGA forms can serve as differentiable functional channels inside the neural model. GradTDDFT also implements implicit differentiation for self-consistent ground-state training and excited-state response training, allowing training objectives to be propagated back to the XC parameters through the SCF and TDDFT maps. In this way, the learned XC energy, SCF potential, response kernel, and excited-state observables are treated within one differentiable training framework.

II. THEORY AND METHODS

A. XC Functional Construction

For a molecular system with density ρ, the Kohn–Sham energy is written as

$$E_\theta[\rho] = T_s[\rho] + E_{\mathrm{ne}}[\rho] + J[\rho] + E_{\mathrm{xc}}^{\theta}[\rho] + E_{\mathrm{nn}}, \tag{1}$$

where T_s is the noninteracting kinetic energy, E_{ne} is the electron–nuclear attraction, J is the Hartree energy, E_{nn} is the nuclear repulsion, and θ denotes the neural XC parameters. In an AO basis, the corresponding generalized eigenvalue problem is

$$\begin{aligned} \mathbf{F}_\theta[\mathbf{D}]\mathbf{C} &= \mathbf{S}\mathbf{C}\boldsymbol{\epsilon}, \\ \mathbf{F}_\theta &= \mathbf{h} + \mathbf{J}[\mathbf{D}] + \mathbf{V}_{\mathrm{xc}}^{\theta}[\mathbf{D}] + \mathbf{V}_{\mathrm{hyb}}^{\theta}[\mathbf{D}]. \end{aligned} \tag{2}$$

where $\mathbf{D}$ is the density matrix and $\mathbf{V}_{\mathrm{hyb}}^{\theta}$ collects nonlocal hybrid contributions.

GradTDDFT uses a coefficient-basis neural hybrid functional. Following the energy-density mixture idea used in GradDFT,[18] the XC energy is written in continuous form as

$$\begin{aligned} E_{\mathrm{xc}}^{\theta} &= \int e_{\mathrm{xc}}^{\theta}(\mathbf{r})\, d\mathbf{r}, \\ e_{\mathrm{xc}}^{\theta}(\mathbf{r}) &= \sum_{m=1}^{M} c_m^{\theta}(\mathbf{z}(\mathbf{r}))e_m^{\mathrm{sl}}(\mathbf{r}) \\ &\quad + c_x^{\theta}(\mathbf{z}(\mathbf{r}))h(\mathbf{r}) + c_2^{\theta}(\mathbf{z}(\mathbf{r}))p(\mathbf{r}). \end{aligned} \tag{3}$$

The energy density is expanded into three channel groups: local or semi-local XC energy densities, a projected local exact-exchange channel, and an optional projected local MP2/PT2 second-order correlation channel. Here $\mathbf{z}(\mathbf{r})$ is a local descriptor vector, c_m^{θ}, c_x^{θ}, and c_2^{θ} are learned pointwise mixing fields, $e_m^{\mathrm{sl}}(\mathbf{r})$ are semi-local energy-density channels, $h(\mathbf{r})$ is the local HF exchange channel density, and $p(\mathbf{r})$ is the local PT2 channel density. Constant coefficients recover conventional hybrid or double-hybrid forms, while learned pointwise coefficients define the neural functional used in GradTDDFT. In numerical calculations, Eq. (3) is evaluated by molecular quadrature.

A general semi-local channel has the form

$$e_m^{\mathrm{sl}}(\mathbf{r}) = \rho(\mathbf{r})\, \varepsilon_m^{\mathrm{sl}}(\rho_\sigma, \nabla\rho_\sigma \cdot \nabla\rho_{\sigma'}, \tau_\sigma), \tag{4}$$

with LDA, GGA, and meta-GGA channels obtained by selecting the corresponding subset of density variables. For example, a spin-unpolarized LDA exchange channel and a GGA exchange channel may be written as

$$\begin{aligned} e_x^{\mathrm{LDA}}(\mathbf{r}) &= -C_x \rho(\mathbf{r})^{4/3}, \\ e_x^{\mathrm{GGA}}(\mathbf{r}) &= e_x^{\mathrm{LDA}}(\mathbf{r}) F_x(s), \\ s &= \frac{|\nabla\rho(\mathbf{r})|}{2(3\pi^2)^{1/3}\rho(\mathbf{r})^{4/3}}, \end{aligned} \tag{5}$$

where $C_x = \frac{3}{4}(3/\pi)^{1/3}$ and F_x is the functional-specific GGA enhancement factor. Through jax-xc[19], different conventional LDA, GGA, and meta-GGA energy-density forms can be inserted into the semi-local channel set without changing the SCF or TDDFT equations.

The HF and MP2/PT2 ingredients enter the coefficient basis through local channel densities whose integrals recover the conventional nonlocal energies. With the spin-density matrix $\gamma_\sigma(\mathbf{r}, \mathbf{r}')$, the exact-exchange channel is

$$\begin{aligned} E_x^{\mathrm{HF}} = \int h(\mathbf{r})\, d\mathbf{r} &= -\frac{1}{2}\sum_{\sigma}\int \frac{|\gamma_\sigma(\mathbf{r}, \mathbf{r}')|^2}{|\mathbf{r} - \mathbf{r}'|}\, d\mathbf{r}\, d\mathbf{r}' \\ &= -\frac{1}{2}\sum_{\sigma}\sum_{ij \in \mathrm{occ}_\sigma} (ij|ji). \end{aligned} \tag{6}$$

For the second-order correlation channel, using spin-orbital indices i,j for occupied orbitals and a,b for virtual orbitals,

$$E_c^{(2)} = \int p(\mathbf{r})\,d\mathbf{r} = -\frac{1}{4}\sum_{ijab}\frac{|\langle ij||ab\rangle|^2}{D_{ij}^{ab}},$$
$$D_{ij}^{ab} = \varepsilon_a + \varepsilon_b - \varepsilon_i - \varepsilon_j, \qquad \langle ij||ab\rangle = (ia|jb) - (ib|ja). \tag{7}$$

Here $(pq|rs)$ denotes a two-electron integral in chemists' notation. The local gauges $h(\mathbf{r})$ and $p(\mathbf{r})$ are not unique; in GradTDDFT they are projected to the molecular grid as local HF and PT2 channel features. Their detailed grid definitions and response binding are given in the Supporting Information.

B. Differentiable SCF and TDDFT

GradTDDFT treats the molecular integrals, basis metadata, and grid quantities as fixed input data $\mathscr{I}$. The density-dependent calculation is written as a differentiable SCF fixed point,

$$\mathbf{D}_\theta^* = \Phi_\theta(\mathbf{D}_\theta^*;\mathscr{I}), \qquad \mathbf{R}_\theta(\mathbf{D}) = \mathbf{D} - \Phi_\theta(\mathbf{D};\mathscr{I}), \tag{8}$$

where Φ_θ denotes Fock construction, generalized diagonalization, occupation assignment, and density reconstruction. For implicit self-consistent differentiation, the converged density satisfies $\mathbf{R}_\theta(\mathbf{D}_\theta^*) = 0$, and its parameter derivative is obtained from

$$\frac{d\mathbf{D}_\theta^*}{d\theta} = -\left(\frac{\partial \mathbf{R}_\theta}{\partial \mathbf{D}}\right)^{-1}\frac{\partial \mathbf{R}_\theta}{\partial \theta}. \tag{9}$$

This implicit SCF relation is the ground-state differential used by the self-consistent training path; alternative fixed-density and finite-SCF differentiation modes are defined in the Supporting Information.

After SCF convergence, excitation energies are obtained from linear-response TDDFT. For occupied orbitals i,j and virtual orbitals a,b, the full Casida equation is

$$\begin{pmatrix}\mathbf{A} & \mathbf{B}\\ \mathbf{B} & \mathbf{A}\end{pmatrix}\begin{pmatrix}\mathbf{X}_I\\ \mathbf{Y}_I\end{pmatrix} = \Omega_I\begin{pmatrix}\mathbf{1} & \mathbf{0}\\ \mathbf{0} & -\mathbf{1}\end{pmatrix}\begin{pmatrix}\mathbf{X}_I\\ \mathbf{Y}_I\end{pmatrix}, \tag{10}$$

where Ω_I is the excitation energy. The response matrices have the schematic form

$$\begin{aligned}A_{ia,jb} &= (\varepsilon_a - \varepsilon_i)\delta_{ij}\delta_{ab} + K_{ia,jb},\\ B_{ia,jb} &= K_{ia,bj},\end{aligned} \tag{11}$$

where K contains Coulomb, XC-kernel, and hybrid-response contributions. The Tamm–Dancoff approximation sets $\mathbf{B} = 0$ and gives

$$\mathbf{A}\mathbf{X}_I = \Omega_I\mathbf{X}_I. \tag{12}$$

The differentiable link between the SCF energy and the TDDFT response is the XC kernel. For the local and semi-local part,

$$f_{\mathrm{xc}}^\theta(\mathbf{r},\mathbf{r}') = \frac{\delta^2 E_{\mathrm{xc}}^\theta}{\delta\rho(\mathbf{r})\delta\rho(\mathbf{r}')}. \tag{13}$$

Thus the SCF potential and the local response kernel are derivatives of the same energy-density model. In GradTDDFT these derivatives are evaluated by automatic differentiation of the pointwise energy density, so derivatives of the learned mixing fields are included in both the potential and the TDDFT kernel. The nonlocal HF response, the PT2 excited-state correction, and optional RI response contractions are specified in the Supporting Information.

For differentiable excited-state training, the response problem is evaluated at the implicit SCF solution $\mathbf{D}_\theta^*$. The TDDFT root derivative therefore contains both the explicit derivative of the response operator and the implicit derivative of the converged density in Eq. (9). In the Davidson response solver, GradTDDFT follows the implicit eigensolver differentiation strategy of Xie, Liu, and Wang,[21] using the converged Ritz amplitudes to reconstruct the root-energy differential rather than differentiating through the full iteration history. For the TDA operator, this takes the Rayleigh-quotient form

$$d\Omega_I = \frac{\mathbf{X}_I^T(d\mathbf{A})\mathbf{X}_I}{\mathbf{X}_I^T\mathbf{X}_I}, \tag{14}$$

with $d\mathbf{A}$ including both direct neural-kernel derivatives and the SCF response from Eq. (9). The corresponding full Casida expression is given in the Supporting Information.

III. IMPLEMENTATION

A. Training Data Flow and Standard Training Loop

GradTDDFT is implemented around the same predictor–loss–optimizer organization used in machine-learning-enhanced DFT workflows such as GradDFT,[18] but the predictor is extended from ground-state DFT to TDDFT response. A training example is represented as

$$\mathscr{D}_k = \left(\mathscr{I}_k, \mathbf{y}_k^{\mathrm{gs}}, \mathbf{y}_k^{\mathrm{es}}, w_k\right), \tag{15}$$

where $\mathscr{I}_k$ contains the fixed numerical input for molecule k, $\mathbf{y}_k^{\mathrm{gs}}$ and $\mathbf{y}_k^{\mathrm{es}}$ denote ground-state and excited-state targets, and w_k is a sample weight. The fixed input includes nuclear data, basis metadata, AO values on the molecular grid, one-electron integrals, two-electron contraction data, initial densities, and optional sources for local HF and PT2 channels. These sources may be cached grid descriptors or auxiliary contraction factors from which state-dependent descriptors are recomputed during SCF. Only the neural functional parameters θ are updated during training.

The training workflow is organized into four operations. First, a PySCF mean-field calculation is converted into a fixed numerical reference containing the molecule, basis metadata, grid information, one- and two-electron data, and optional local HF and PT2 descriptors. Second, the neural XC functional is assembled by choosing the semilocal energy-density channels and by enabling the HF and PT2 channels required by the calculation. The semilocal channels define the local energy-density basis, whereas the HF and PT2 options control additional functional components and the local descriptors that condition the coefficient model.

Third, the training mode is selected. Fixed-density training evaluates the objective on supplied densities, while self-consistent training first solves the differentiable SCF problem before evaluating the observable. The self-consistent path can use either explicit differentiation through the SCF trace or an implicit fixed-point gradient. Excited-state training uses the same molecular datum but adds TDA or Casida response targets and the selected response backend, such as the RIS approximation used for larger response calculations. Finally, ground-state and excited-state targets are packaged with the fixed molecular reference and passed to the optimizer. The two training tasks share the same data-flow abstraction, but the reported ground-state and excited-state models are optimized in separate runs.

For a batch $\mathscr{B}$, the differentiable predictor maps θ and the fixed molecular inputs to observables,

$$\widehat{\mathbf{y}}_k(\theta) = \mathscr{O}\left[\mathscr{T}_\theta(\mathscr{I}_k)\right], \qquad k \in \mathscr{B}. \tag{16}$$

Here $\mathscr{T}_\theta$ denotes the selected forward electronic structure calculation: fixed-density evaluation, self-consistent SCF, or SCF followed by TDA/Casida response. The observable map $\mathscr{O}$ extracts the quantities used by the current objective, such as total energies, orbital diagnostics, excitation energies, oscillator strengths, or broadened spectra. The standard training iteration is

$$\mathscr{L}_{\mathscr{B}}(\theta) = \frac{1}{\sum_{k\in\mathscr{B}} w_k} \sum_{k\in\mathscr{B}} w_k\, \ell\left(\widehat{\mathbf{y}}_k(\theta), \mathbf{y}_k\right), \tag{17}$$

$$\theta_{t+1} = \mathscr{U}\left(\theta_t, \nabla_\theta \mathscr{L}_{\mathscr{B}}(\theta_t)\right), \tag{18}$$

where $\mathscr{U}$ is the optimizer update. In practice the loss and its gradient are evaluated by reverse-mode automatic differentiation with respect to θ, followed by finite-value checks, gradient sanitization, gradient-norm logging, and parameter-update logging. Ground-state and excited-state targets share this data representation, but the calculations reported here optimize ground-state and excited-state objectives in separate runs.

The SCF part of $\mathscr{T}_\theta$ can be differentiated either through a finite sequence of SCF iterations or through the fixed-point adjoint described in Section II. The TDDFT part uses a Davidson forward solve for the requested low-lying TDA or Casida roots. During training, the reverse pass uses the implicit eigensolver differentiation of Xie, Liu, and Wang[21] applied to the converged Ritz pairs, so the gradient is not propagated through the full Davidson iteration history.

B. Component Gradient Evaluation

The neural functional separates neural-network inputs from the functional channels that enter the energy density. At grid point g, the network input descriptor is written as

$$\mathbf{x}_g = \left(\mathbf{u}_g, d_g^{\mathrm{sl}}, \eta_g^{\mathrm{HF}}, \eta_g^{\mathrm{PT2}}\right), \qquad \mathbf{c}_g^\theta = \mathscr{N}_\theta(\mathbf{x}_g), \tag{19}$$

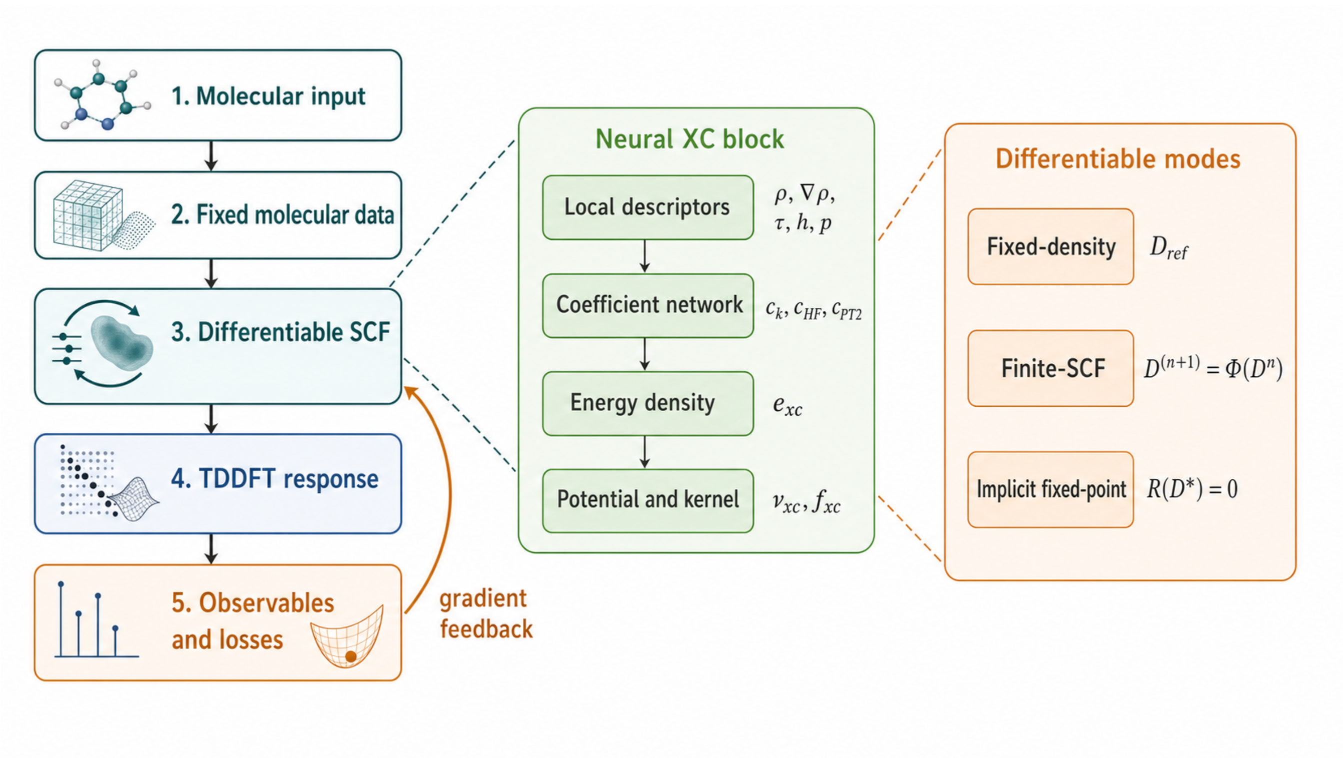

FIG. 1. Training data flow in GradTDDFT. Fixed molecular inputs are passed to a differentiable SCF or SCF–TDDFT predictor; scalar losses are formed from selected observables, and reverse-mode differentiation supplies gradients with respect to the neural XC parameters.

where $\mathbf{u}_g$ contains local density variables, d_g^{sl} denotes the optional semilocal descriptor used by the coefficient model, and η_g^{HF} and η_g^{PT2} denote optional HF and PT2 input descriptors. These descriptors determine the neural mixing field but are not themselves the definition of the XC energy channel. For grid weight w_g, the functional channel basis is

$$\mathbf{b}_g = \left(e_{1,g}^{\mathrm{sl}}, \ldots, e_{M,g}^{\mathrm{sl}}, e_g^{\mathrm{HF}}, e_g^{\mathrm{PT2}}\right), \qquad e_{\mathrm{xc},g}^{\theta} = \mathbf{c}_g^{\theta} \cdot \mathbf{b}_g, \tag{20}$$

where $e_{m,g}^{\mathrm{sl}}$ are local or semilocal energy-density channels, while e_g^{HF} and e_g^{PT2} are the local representations of the HF and PT2 functional components. The same underlying local exchange or pair-gauge quantities may be used both in $\mathbf{x}_g$ and in $\mathbf{b}_g$, but their roles are different: the former conditions the neural coefficients, whereas the latter contributes to the energy density. The direct parameter gradient at a fixed electronic state is therefore assembled channel by channel as

$$\frac{\partial E_{\mathrm{xc}}^{\theta}}{\partial \theta} = \sum_g w_g \mathbf{b}_g^{\mathrm{T}} \frac{\partial \mathbf{c}_g^{\theta}}{\partial \theta}. \tag{21}$$

At a fixed SCF or TDDFT state, Eq. (21) is the direct parameter-gradient path for all channels. In self-consistent calculations the input descriptors and channel values are not restricted to their initial reference values; when the required source tensors are available, the HF and PT2 quantities are rebuilt from the current density and orbital state.

For SCF and TDDFT response, GradTDDFT also needs derivatives with respect to the density variables. Let $\mathbf{u}_g = (\rho_g, \nabla\rho_g, \tau_g, \nabla^2\rho_g)$ denote the active local variables for the selected LDA, GGA, or meta-GGA channel. The local part of the XC potential and the local response action are computed as

$$V_{\mathrm{loc}}^{\theta} = \sum_g w_g \frac{\partial e_{\mathrm{xc},g}^{\theta}}{\partial \mathbf{u}_g} \frac{\partial \mathbf{u}_g}{\partial P}, \tag{22}$$

$$K_{\mathrm{loc}}^{\theta}[\Delta P] = \frac{\partial V_{\mathrm{loc}}^{\theta}}{\partial P}[\Delta P]. \tag{23}$$

For traditional semilocal functionals, the energy density and its response tensor are evaluated through JAX-XC.[19] For neural functionals, the same derivatives are generated from the pointwise energy-density map in Eq. (20). The implementation uses

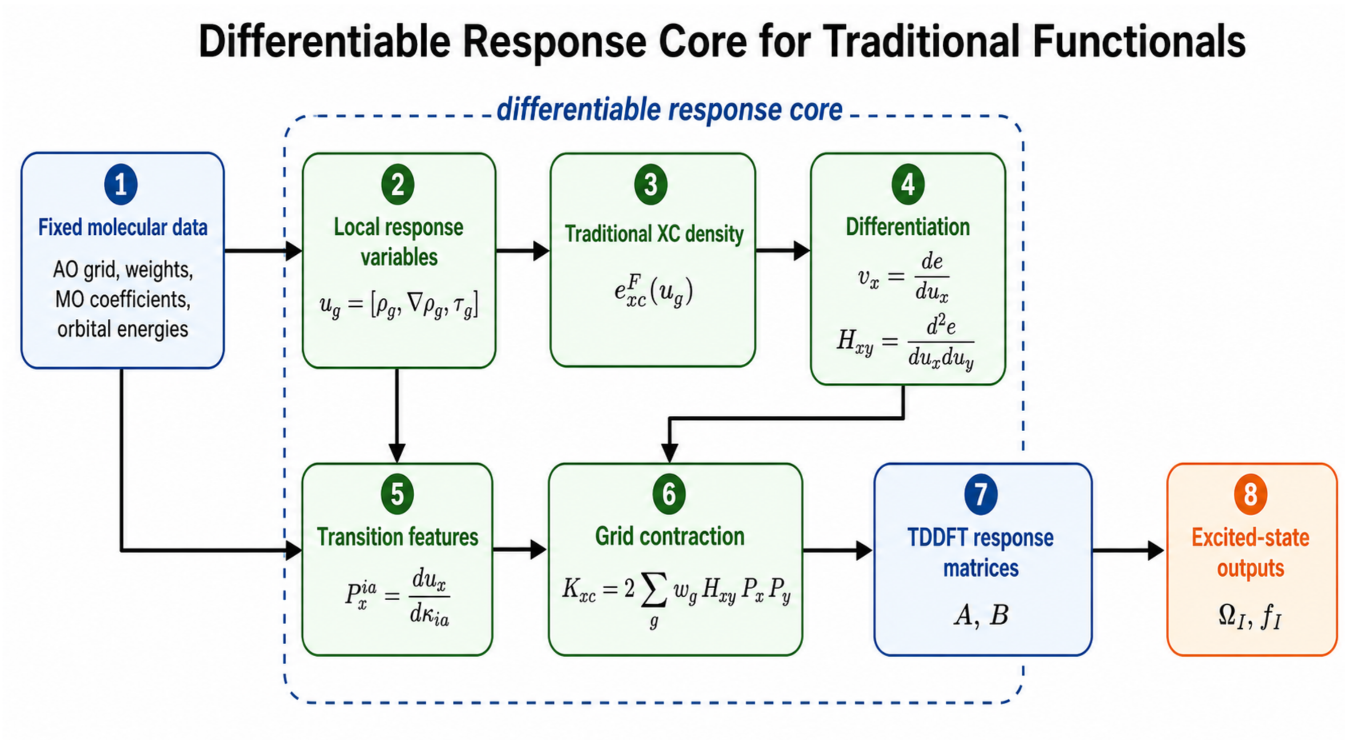

FIG. 2. Differentiable response-core construction. Fixed molecular inputs define local density and transition-density variables on the grid. Differentiation of the same XC energy-density map produces the SCF potential and the local response action used in TDDFT.

Hessian-vector products for response actions when possible, avoiding explicit construction of a dense grid–grid kernel. Figure 2 summarizes this response core for semilocal channels.

The HF information therefore appears in two places. As a neural input, η_g^{HF} conditions the coefficient field $\mathbf{c}_g^{\theta}$. As a functional component, e_g^{HF} contributes an exchange channel to the energy density and determines the effective HF mixing used in the nonlocal exchange contractions. During self-consistent forward execution, these local HF quantities are evaluated from the current density matrix and orbital state when local exchange factors are available, so the HF channel follows the SCF update. For derivative construction, GradTDDFT keeps the local-HF descriptor values and auxiliary exchange factors fixed inside the local response tensor; gradients with respect to θ still pass through the learned HF coefficient field. The resulting mixing field is either averaged to the scalar exchange fraction used in the TDDFT response operator or contracted back to an explicit exchange-like Fock contribution. The omitted part is the full local-HF second response term associated with differentiating the local exchange descriptor itself in the TDDFT kernel.

The PT2 information is organized in the same two-role manner. η_g^{PT2} may be used as a neural input descriptor, while e_g^{PT2} is the PT2 channel in the energy-density basis. In ground-state training it contributes to Eq. (20) and to the local potential derivatives in Eqs. (22) and (23). In the excited-state calculations reported here, the learned PT2 coefficient is converted to a density-weighted scalar amplitude and applied to a state-specific second-order correction after the singles response roots are obtained. This keeps the neural PT2 supervision connected to the differentiable TDDFT predictor while avoiding an additional local Hessian term for the second-order correction itself.

IV. COMPUTATIONAL EVALUATION

The computational evaluation begins with a numerical validation against PySCF and then turns to model-capability tests. The validation module checks that GradTDDFT reproduces conventional TDA and full TDDFT excitation energies, oscillator strengths, and response matrices before neural parameters are introduced. The capability tests then ask whether differentiable neural exchange-correlation functionals can learn ground-state surfaces, excited-state response, and molecular ground-state and vertical-excitation targets within a shared SCF–TDDFT implementation. These model tests are divided into two blocks: bond-dissociation curves covering ground-state surfaces and first excited-state response, and separate ground-state and excited-state training on a 50-molecule QM9 subset.

Across all experiments, the recorded computational state includes the geometry source, charge and spin multiplicity, basis set,

TABLE I. B3LYP/def2-SVP S_1 excitation energies and oscillator strengths from matched PySCF and GradTDDFT calculations. Each entry reports Ω_1 in eV followed by f_1.

Molecule	Solver	PySCF	GradTDDFT
H_2O	TDA	$7.6 / 1.8 \times 10^{-2}$	$7.6 / 1.8 \times 10^{-2}$
H_2O	TDDFT	$7.6 / 1.8 \times 10^{-2}$	$7.6 / 1.8 \times 10^{-2}$
CO	TDA	$8.7 / 1.1 \times 10^{-1}$	$8.7 / 1.1 \times 10^{-1}$
CO	TDDFT	$8.5 / 8.2 \times 10^{-2}$	$8.5 / 8.2 \times 10^{-2}$
N_2	TDA	$9.5 / 4.5 \times 10^{-31}$	$9.5 / 3.4 \times 10^{-31}$
N_2	TDDFT	$9.4 / 4.7 \times 10^{-18}$	$9.4 / 7.2 \times 10^{-25}$
C_2H_4	TDA	$8.3 / 5.4 \times 10^{-29}$	$8.3 / 6.0 \times 10^{-29}$
C_2H_4	TDDFT	$8.1 / 3.7 \times 10^{-1}$	$8.1 / 3.7 \times 10^{-1}$
CH_2O	TDA	$4.0 / 1.9 \times 10^{-29}$	$4.0 / 1.8 \times 10^{-29}$
CH_2O	TDDFT	$4.0 / 2.2 \times 10^{-25}$	$4.0 / 5.4 \times 10^{-29}$
C_6H_6	TDA	$5.5 / 3.7 \times 10^{-9}$	$5.5 / 3.7 \times 10^{-9}$
C_6H_6	TDDFT	$5.5 / 4.9 \times 10^{-9}$	$5.5 / 4.9 \times 10^{-9}$

quadrature grid, SCF convergence criteria, TDDFT solver, excited-state treatment, backend/device, training seed, and reference method. These records are part of the benchmark definition because small changes in grids, spin treatment, or SCF convergence can otherwise obscure whether an observed error comes from the learned functional or from the numerical setup.

A. Numerical Validation against PySCF

The first evaluation module isolates the TDDFT response construction before any neural functional parameters are trained. In the conventional-functional limit, GradTDDFT evaluates the exchange-correlation energy density as a differentiable function of the density descriptors and obtains both the potential and the local response tensor by automatic differentiation,

$$E_{\mathrm{xc}} \longrightarrow v_{\mathrm{xc}} = \frac{\delta E_{\mathrm{xc}}}{\delta \rho} \longrightarrow f_{\mathrm{xc}} = \frac{\delta v_{\mathrm{xc}}}{\delta \rho}. \tag{24}$$

The resulting tensor in Eq. (24) is contracted with quadrature weights and AO basis values, transformed to the occupied–virtual orbital basis, and inserted into the TDA or Casida response matrices. Matched PySCF calculations provide independent references for the same geometry, functional, basis, grid, SCF solution, and response equations.

The validation matrix contains H_2O, CO, N_2, C_2H_4, formaldehyde, benzene, and OH with PBE, B3LYP, and PBE0 in def2-SVP and aug-cc-pVDZ basis sets. Restricted KS references are used for the closed-shell molecules, while unrestricted KS references are used for OH. For each matched setup, TDA and full TDDFT are evaluated for the first five singlet and triplet states where the reference calculation is defined.

For the PBE/def2-SVP subset reported here, all 26 calculations completed successfully, giving 130 per-state records. The direct GradTDDFT–PySCF comparison contains 12 restricted-singlet tasks, corresponding to 60 excitation states across TDA and full TDDFT. The excitation-energy mean absolute differences relative to PySCF range from 5.86×10^{-14} to 2.87×10^{-10} eV, with maximum absolute state errors below 5.47×10^{-10} eV. Oscillator-strength mean absolute differences range from 1.86×10^{-22} to 1.29×10^{-7}. For response matrices small enough to store explicitly in this audit, the RMS differences in the assembled A and B matrices are 5.65×10^{-15} to 8.26×10^{-15}.

This validation establishes the numerical equivalence of the differentiable TDDFT path and matched PySCF calculations for traditional functionals. The following sections therefore evaluate model behavior rather than basic response-matrix assembly.

B. Ground- and Excited-State Dissociation Curves

The first model-capability benchmark evaluates one-dimensional bond dissociation curves for H_2, H_2^+, and N_2, with first excited-state scans for H_2 and N_2. The H_2/H_2^+ pair follows the Kohn–Sham regularizer benchmark of Li *et al.*: H_2 is a compact bond-breaking and strong-correlation test, whereas H_2^+ isolates one-electron self-interaction and open-shell SCF stability.[7] N_2 extends this setting to a stretched multiple bond; recent high-level spectroscopic calculations treat molecular nitrogen as a dense multi-state problem, reporting potential-energy curves and transition dipole moments for the 51 lowest electronic states over 1.35–10 bohr and multiple N + N dissociation limits.[22] The three systems therefore separate self-interaction, two-electron bond breaking, and stretched triple-bond near-degeneracy within a small set of scans. The ground-state curves are labeled by the corresponding diatomic spectroscopic terms, $X\,^1\Sigma_g^+$ for H_2 and N_2 and $X\,^2\Sigma_g^+$ for H_2^+. The excited-state curves are labeled as H_2 $B\,^1\Sigma_u^+$ and N_2 $A\,^1\Pi_g$; the latter follows the large-CAS state assignment used in the N_2 reference data.[22]

The results are organized around the physically meaningful channel for each system. For the one-electron H_2^+ ion, the MP2/PT2 doubles space is empty, so a PT2 correction is identically zero and does not provide a useful functional test. H_2^+ is therefore used to compare implicit-SCF training with fixed-density training. For H_2 and N_2, the implicit-SCF protocol is used throughout, and the PT2/no-PT2 comparison tests how strongly the second-order channel improves the learned functional along bond dissociation. The H_2 and N_2 first excited-state dissociation curves apply the same PT2-channel test to the TDDFT response calculation. The curve-plus-error layout, training markers, and chemical accuracy band are used as direct diagnostics, with convergence success, cycle count, and failure rate recorded at every geometry.

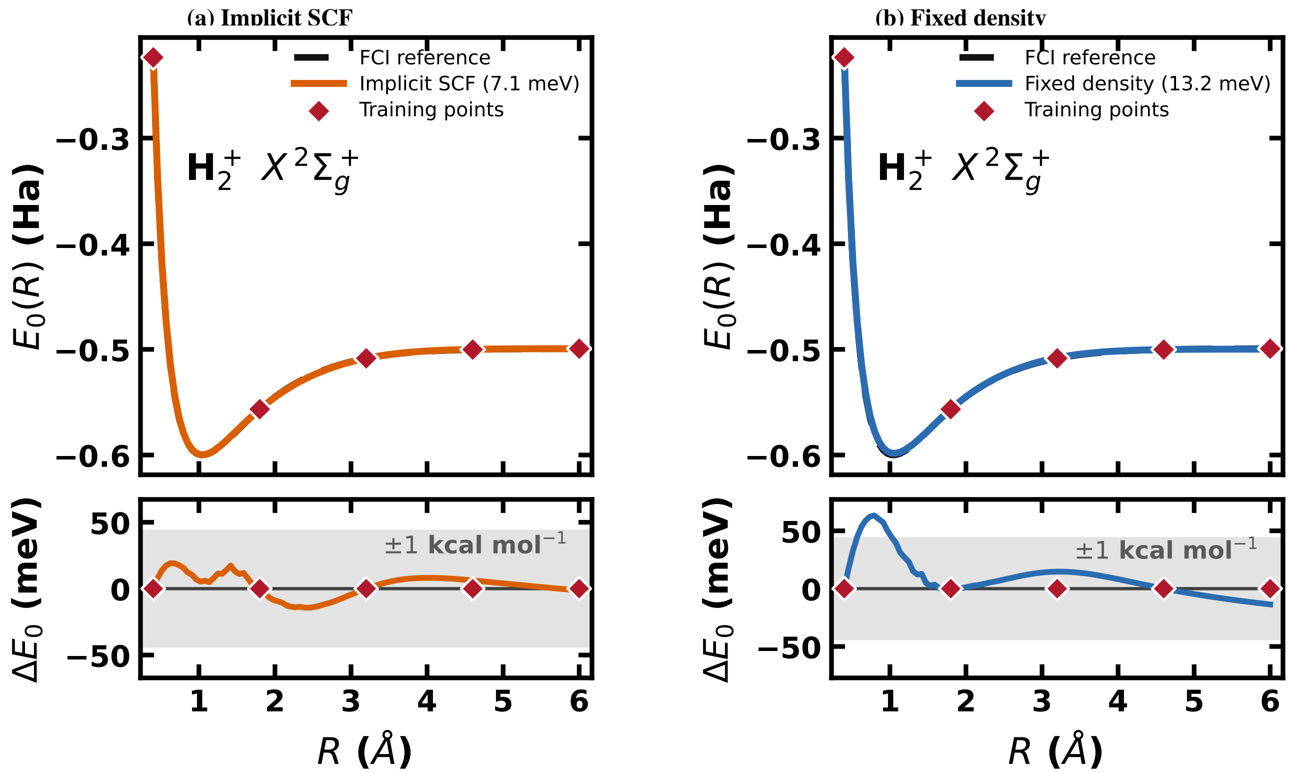

FIG. 3. H_2^+ $X\,^2\Sigma_g^+$ ground-state dissociation curves for the HFX-channel neural functional under implicit-SCF and fixed-density training. In each panel, the upper subpanel compares the learned curve with the one-electron FCI reference values, while the lower subpanel shows the signed error $\Delta E_0 = E_{0,\mathrm{model}} - E_{0,\mathrm{ref}}$. Red diamonds denote the five training geometries, and the gray band marks ± 1 kcal mol^{-1}. The implicit-SCF model gives a dense-curve mean absolute error of 7.1 meV and a maximum absolute error of 19.2 meV. Fixed-density training gives corresponding errors of 13.2 meV and 63.2 meV.

C. Semi-local Channel Composition Tests

The dissociation benchmarks above vary the implicit-SCF treatment and the second-order PT2 channel. A complementary training test asks a narrower question: under a fixed `noPT2+HFX` implicit-SCF protocol, how strongly does the semi-local channel basis affect the N_2 ground-state and first-excited-state dissociation curves? In this comparison, the MP2/PT2 channel is disabled, the local exact-exchange descriptor h_g remains active through $c_x^\theta(\mathbf{z}_g)h_g$, and only the set of semi-local energy densities $\{e_m^{\mathrm{sl}}\}$ in Eq. (4) is changed. The neural architecture, optimizer, training geometries, quadrature grid, and implicit differentiation of the SCF–TDDFT workflow are kept fixed.

We refer to these models as semi-local channel-composition variants. The functional labels identify the JAX-XC energy-density components used to build the coefficient basis; they do not introduce fixed parent-functional mixing fractions. Because this benchmark uses the `noPT2` setting, no $c_2^\theta(\mathbf{z}_g)p_g$ contribution is included. The comparison therefore isolates how LDA, GGA, and meta-GGA semi-local priors interact with the learned local HFX channel in the coefficient-basis form of Eq. (3).

Each variant is trained and evaluated with the same N_2 geometries, reference labels, basis/grid settings, and response solver.

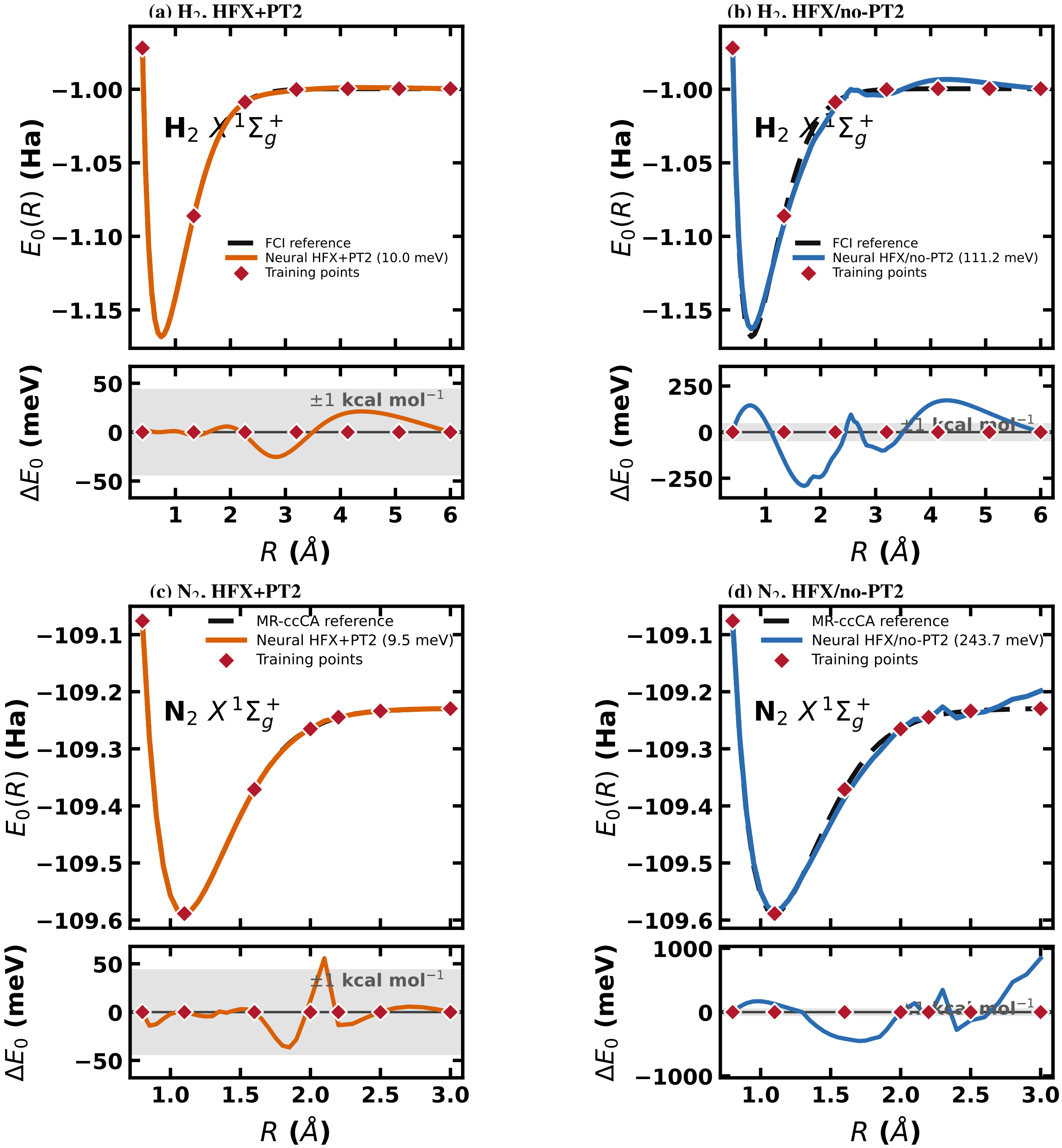

FIG. 4. H_2 and N_2 $X\,^1\Sigma_g^+$ ground-state dissociation curves for neural functionals trained with and without the PT2 channel. In each panel, the upper subpanel compares the learned curve with the reference values, while the lower subpanel shows the signed error $\Delta E_0 = E_{0,\mathrm{model}} - E_{0,\mathrm{ref}}$. Red diamonds denote the seven training geometries, and the gray band marks ± 1 kcal mol^{-1}. For H_2, the HFX+PT2 model gives a dense-curve mean absolute error of 10.0 meV and a maximum absolute error of 25.4 meV, while removing the PT2 channel increases the corresponding errors to 111.2 meV and 292.1 meV. For N_2, the HFX+PT2 model gives corresponding errors of 9.5 meV and 55.7 meV, whereas the HFX/no-PT2 model gives 243.7 meV and 844.0 meV.

The ground-state comparison uses $E_0(R)$; the excited-state comparison uses the first excitation energy $\Omega_1(R)$ and the total first-excited-state curve $E_1(R) = E_0(R) + \Omega_1(R)$. Since HFX is enabled and PT2 is disabled in all rows of Table II, differences across the three variants are attributed to the semi-local channel basis rather than to changes in the nonlocal response ingredients.

Within the same dissociation benchmark, training on near-equilibrium geometries is compared with training on the full curve. The near-equilibrium regime tests extrapolation into stretched bonds, whereas the full-curve regime tests interpolation across

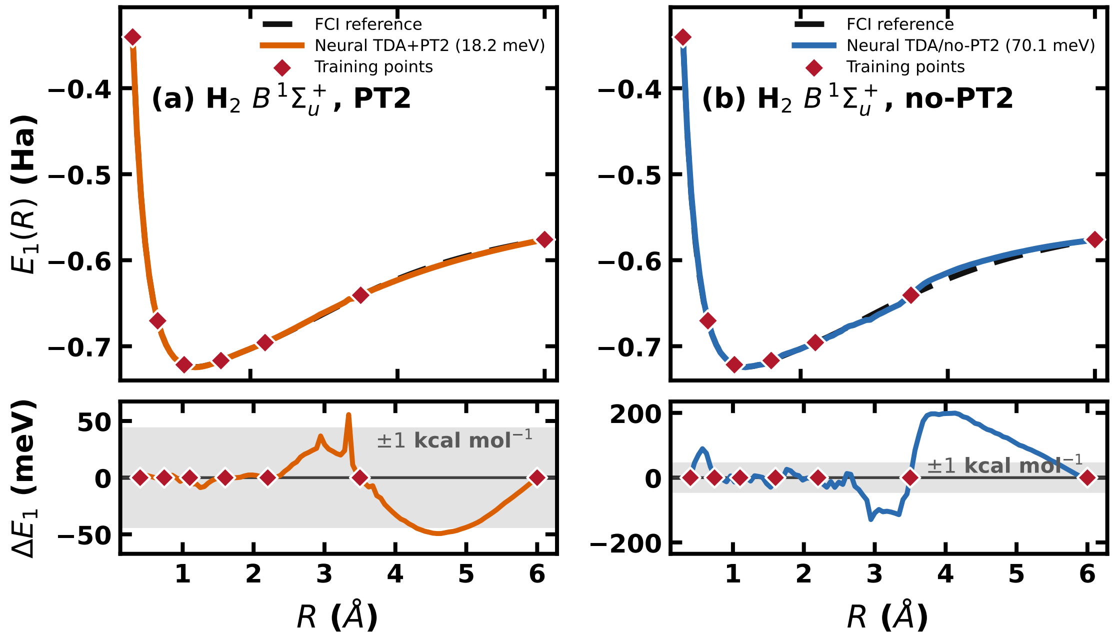

FIG. 5. H_2 $B\,^1\Sigma_u^+$ excited-state dissociation curve for neural TDA calculations with and without the PT2 channel. The plotted quantity is the total excited-state energy $E_1(R) = E_0(R) + \Omega_1(R)$. In each panel, the upper subpanel compares the learned curve with the FCI reference values, while the lower subpanel shows the signed error $\Delta E_1 = E_{1,\mathrm{model}} - E_{1,\mathrm{ref}}$. Red diamonds denote the training geometries, and the gray band marks ± 1 kcal mol^{-1}. The PT2-channel model gives a dense-curve mean absolute error of 18.2 meV and a maximum absolute error of 55.6 meV, whereas the updated no-PT2 model gives corresponding errors of 70.1 meV and 199.3 meV.

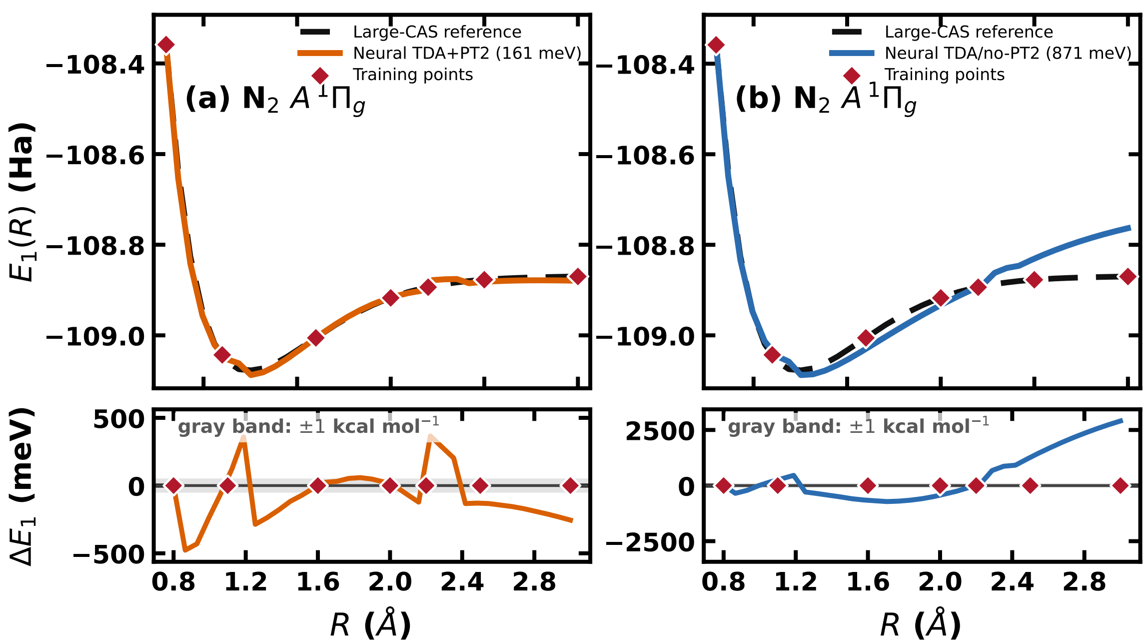

FIG. 6. N_2 $A\,^1\Pi_g$ excited-state dissociation curves for neural TDA calculations with and without the PT2 channel. The plotted quantity is the total excited-state energy $E_1(R) = E_0(R) + \Omega_1(R)$, using the Hammami *et al.* large-CAS curve as the reference.[22] In each panel, the upper subpanel compares the learned curve with the reference values, while the lower subpanel shows the signed error $\Delta E_1 = E_{1,\mathrm{model}} - E_{1,\mathrm{ref}}$. Red diamonds denote the training geometries, and the gray band marks ± 1 kcal mol^{-1}. The PT2-channel model gives a dense-curve mean absolute error of 0.161 eV and a maximum absolute error of 0.477 eV, while removing the PT2 channel increases the corresponding errors to 0.871 eV and 2.905 eV.

TABLE II. Semi-local channel-composition benchmark for N_2 dissociation. The MP2/PT2 channel is disabled, the local HFX channel is enabled, and all variants use the implicit-SCF training path. Only the semi-local basis $\{e_m^{\mathrm{sl}}\}$ is changed.

Variant	Semi-local channel basis	Dissociation diagnostic
`lda_vwn_rpa`	`lda_x` + `lda_c_vwn_rpa`	Baseline for $E_0(R)$ and $E_1(R)$
`gga_pbe`	`gga_x_pbe` + `gga_c_pbe`	GGA exchange/correlation prior
`mgga_r2scan`	`mgga_x_r2scan` + `mgga_c_r2scan`	Meta-GGA τ-dependent prior

both equilibrium and dissociation regions. This mirrors the GradDFT-style potential-energy-surface logic while adding explicit SCF convergence diagnostics.

The excited-state scans test whether the same differentiable SCF–TDDFT workflow can support response calculations along a dissociation coordinate. For each bond length, the calculation records the ground-state energy $E_0(R)$, the first excitation energy $\Omega_1(R)$, and the first excited-state energy

$$E_1(R) = E_0(R) + \Omega_1(R). \tag{25}$$

Reference curves for Eq. (25) are generated with a fixed basis, spin sector, and active space across the scan so that errors reflect the learned functional and response treatment rather than changing reference definitions.

The comparison includes TDA, full TDDFT, and neural-XC TDDFT under the same orbital and response settings. Neural objectives are grouped into ground-state-only and excited-state-only losses. This separation tests whether fitting $E_0(R)$ alone yields a useful response kernel and whether direct S_1 supervision perturbs the ground-state surface. The key diagnostic is the compatibility between the learned SCF fixed point and the learned response kernel along the entire bond scan under the corresponding separate objective.

D. Separate Ground- and Excited-State Training on 50 QM9 Molecules

The second model-capability benchmark evaluates separate ground-state and excited-state training on a 50-molecule subset of QM9.[23] The subset contains small organic molecules at fixed ground-state geometries and is used to test whether a neural functional can learn both total-energy and vertical-excitation targets beyond one-dimensional bond scans. The implementation remains self-consistent in both protocols: neural parameters affect the SCF solution, TDDFT response matrices, excitation energies, and oscillator strengths.

The primary targets are ground-state total energies and the lowest vertical excitation energies. Oscillator strengths and state ordering are retained as response-quality diagnostics. Molecule-level splits, random seeds, charge and spin assignments, basis sets, grids, and reference labels are recorded with the training data. Results are reported separately for ground-state-only and excited-state-only objectives, together with molecular holdout errors within the 50-molecule subset, response solver convergence, and SCF failure rate. This block tests whether the differentiable SCF–TDDFT training path can move from controlled dissociation scans to chemically diverse small organic molecules under the two separate training protocols.

V. DISCUSSION

GradTDDFT is designed to make the response kernel part of the trainable computational object rather than an after-the-fact diagnostic. This distinction is important for neural exchange-correlation functionals because excitation energies depend not only on the converged KS orbitals and eigenvalue gaps, but also on the derivative structure of the learned energy density. The energy-consistent local response construction therefore enforces consistency among the neural XC energy, the XC potential, and the local f_{xc} kernel. This consistency is a central requirement when the same functional is used in both SCF optimization and TDDFT response.

The evaluation first separates numerical correctness from model capability. Matched PySCF calculations test whether the differentiable implementation reproduces conventional TDA and full TDDFT in the non-neural limit. After this check, the model-capability benchmarks separate sources of error that are often mixed in neural-functional studies. H_2, H_2^+, and N_2 ground-state dissociation curves test whether the learned functional remains stable inside an SCF loop across equilibrium and stretched geometries. H_2 and H_2^+ first excited-state dissociation curves then test whether the same functional gives compatible ground-state surfaces and response kernels. The 50-molecule QM9 ground- and excited-state training set tests whether the differentiable SCF–TDDFT training graph transfers from controlled bond scans to chemically diverse small organic molecules under separate total-energy and vertical-excitation objectives. This separation is necessary because a low energy error, a stable SCF solution, and an accurate excitation spectrum are related but not equivalent requirements.

This is also why the present training protocol keeps ground-state and excited-state supervision separate. A joint loss is mathematically straightforward in a differentiable SCF–TDDFT graph, but it is not a robust practical objective for the current neural functional. The ground-state loss primarily drives the self-consistent density, orbital energies, and total-energy surface, whereas the excited-state loss drives the response kernel, excitation energies, oscillator strengths, and state ordering. When both are optimized at the same time, their gradients can compete: updates that improve the ground-state surface may degrade the response kernel, and updates that fit a target excitation can destabilize the SCF fixed point or damage the ground-state energy landscape. In our training design, separating the objectives makes these failure modes observable rather than hiding them inside an unstable multi-target optimization.

The nonlocal channels define a separate theoretical boundary. For semilocal channels, automatic differentiation of the local energy density gives a direct route to a consistent response kernel. For the HF channel, the current excited-state path uses the learned local exchange field to define a scalar hybrid fraction and then applies the ordinary exact-exchange response matrix. For the MP2/PT2 channel, the response problem remains a singles TDA/Casida problem and the learned correlation coefficient scales a root-specific CIS(D)-type doubles correction. This separation is useful for charge-transfer states, double-excitation character, and open-shell systems where the chosen excited-state correction can change the qualitative state ordering.

Several extensions remain natural next steps. A future direct orbital-Hessian treatment of the MP2/PT2 channel would provide a more direct connection between dynamical correlation descriptors and excited-state response. A JAX-native integral path would reduce the separation between the fixed molecular data layer and the differentiable SCF layer, although efficient production use still requires optimized J/K construction and memory management. General unrestricted differentiable SCF would broaden the training targets to radicals, charged species, and spin-polarized response calculations. Larger basis sets and larger molecules will further test whether the response-kernel construction remains practical when the orbital and quadrature spaces grow.

VI. CONCLUSIONS

We introduced GradTDDFT, a differentiable TDDFT framework for training neural XC functionals with excited-state observables. The framework combines PySCF molecular data with JAX implementations of SCF, functional evaluation, response-kernel construction, and TDA/Casida observables.

GradTDDFT connects neural functional parameters to ground-state energies, excitation energies, and oscillator strengths in one computational graph. This provides a practical platform for testing neural functionals beyond ground-state fitting, including scalar hybrid exchange response and CIS(D)-type PT2 corrections in excited-state calculations.

ACKNOWLEDGMENTS

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SUPPLEMENTARY MATERIAL

The supplementary material contains the following files.

- Detailed theory and response-kernel derivations
- Computational input records for the molecular benchmarks
- Reference geometries, grids, basis choices, and convergence settings used in the reported calculations
- High-resolution versions of the figures in the main text

CODE AVAILABILITY

GradTDDFT is available under the Apache 2.0 license on GitHub. The released code contains the research implementation and supporting materials for this manuscript. Numerical values should be interpreted together with the recorded geometry, basis, grid, convergence, backend, and reference settings.

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